

effective software. Because of its growing popularity, Python is receiving more funding for development, which expands its capabilities and ecosystem.

Let's start coding with Python

Getting started with Python is a straightforward process. To install it, follow the steps given here. For our coding setup, we will use the following tools:

- Visual Studio Code
- Python 3
- GitHub
- Rapid API

Download Python: Visit the official Python website at <https://www.python.org/>. Navigate to the *Downloads* section, and choose the appropriate installer for your operating system (Windows, macOS, or Linux). As of now, Python 3.x is the recommended version for beginners.

Install Python:

- **Windows:** Double-click the downloaded installer, and follow the installation prompts. Make sure to check the box that says 'Add Python x.x to PATH' during installation to easily access Python from the command line.
- **macOS:** Run the downloaded installer and follow the installation instructions. Python should already be installed on most macOS systems, but you may need to update it to the latest version.
- **Linux:** Python is usually pre-installed on most Linux distributions. Use your package manager to install it if it's not already available. For example, on Ubuntu, you can install Python by running `sudo apt install python3`.

Verify installation: After installation, open a terminal or command prompt and type `python --version` (or `python3 --version` depending on your system) to check if Python is installed correctly. You should see the installed Python version on the screen.

Interactive mode: Python comes with an interactive mode interpreter, which allows you to execute Python commands line by line. To start the Python interpreter, simply type `python` (or `python3`) in your terminal or command prompt and press *Enter*. You should see the Python prompt (`>>>`), indicating that you're in interactive mode.

Writing your first Python program:

- Launch VS Code (a text editor such as Notepad on Windows, TextEdit on macOS, or any code editor of your choice).
- Write your Python code. For example, you can start with a simple 'Hello, World!' program:

```
print("Hello, World!")
```

- Save the file with a `.py` extension (e.g., `hello.py`) in a directory of your choice.

Running your Python program:

- Open a terminal or command prompt (*Ctrl + Shift + `* & *Cmd + Shift + `*)
- Navigate to the directory where you saved your Python file using the `cd` command.
- Type `python hello.py` (or `python3 hello.py`) and press *Enter*. You should see 'Hello, World!' on the screen. Congratulations! You've successfully installed Python and written your first Python program. You can continue exploring Python by learning its syntax, data structures, and various libraries and frameworks for different purposes, such as web development, data analysis, and machine learning.

Community support and resources for Python

- **Python open source community:** Joining the Python open source community is a fantastic way to connect with fellow developers, seek guidance, and contribute to various projects. Websites like GitHub provide a platform for collaboration and exploration of open source Python projects.
- **Articles on Medium:** Medium is a treasure trove of Python tutorials, articles, and tips shared by experienced developers and educators. Utilise the search feature to find articles tailored to your learning needs and level of expertise.
- **GeeksforGeeks Python tutorial:** GeeksforGeeks offers a comprehensive Python tutorial covering a wide range of topics, from basics to advanced concepts. The step-by-step explanations and code examples make it an invaluable resource for beginners.

Essential Python modules

- **Flask:** Flask is a lightweight and flexible web framework for Python, perfect for building small to medium-sized web applications with ease.
- **FastAPI:** FastAPI is a modern, fast (high-performance) web framework for building APIs with Python 3.7+ based on standard Python type hints.
- **Django:** Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design.
- **Requests:** Requests is a simple yet powerful HTTP library for Python, making it easy to send HTTP requests and handle responses.
- **Psycopg2:** Psycopg2 is a PostgreSQL adapter for Python, enabling seamless interaction with PostgreSQL databases from Python applications.
- **Pyrogram:** Pyrogram is a Python library that provides a convenient way to interact with the Telegram API, facilitating the development of Telegram bots and applications.